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Environmental Studies

Q.1 Give definition, scope and importance of environmental studies.

→ Environmental study is the academic field which systematically studies every issue that affects an organism.

→ Scope of Environmental Studies is in numerous fields:

(1) conservation & management of natural resources (like forest resources, water resources, etc.)
(2) conservation of biodiversities (like conservation of genetic diversity, species diversity, ecosystem diversity, landscape diversity etc.)
(3) control of environmental pollutions (like air pollution, water pollution, soil pollution, electronic-waste pollution, e-pollution etc.)
(4) control of human population
(5) Replacement of development (like green revolution, urbanisation), economic growth) with sustainable developme

→ Importance of Environmental Studies

(1) It directs attention towards the unlimited exploitation of environment by humans for greed or for the sake of development.
(2) It generates concern for the changing environment, population explosion & throws light on the methods of solution.
(3) It helps to understand different food chains to find ways & means to maintain ecological balance.
(4) It helps in maintenance of healthy life. Through improved health of people, economic productivity gets increased.
(5) It imparts knowledge about conservation of energy and reducing material dependence

(a) by refusing to purchase things which are harming our environment
(b) by reusing a product number of times
(c) by motivating recycling of recyclable products

(6) It helps in developing social responsibility towards protection of environment & control of environmental pollution.

(7) It helps in appreciating and enjoying nature and working towards protection of environment & control of environmental pollution.

~~(8) It helps in app~~

Q2 Describe the objectives of environmental studies.

The objective is to help social groups & individuals to

(i) S → skills
acquire the skills for identifying & solving environmental problems

(ii) P → protection qualities
participate in improvement & protection of environment

(iii) E → evaluation abilities
develop the ability to evaluate measures for the improvement and protection of environment.

(iv) A → attitude of concern
acquire an attitude of concern for the environment

(v) K → knowledge
gain a variety of experiences & acquire a basic understanding and knowledge about the environment and its allied problems.

To sum up, the objective of environmental studies is to develop a world in which persons are aware of & concerned about the environment & the problems associated with it, and committed to work individually as well as collectively towards solutions of current problems & prevention of future problems

Q3 Define food chain and food web with examples.

Food chain

Food chain is a feeding hierarchy in which organisms in an ecosystem are grouped into nutritional (trophic) levels and are shown in succession to represent the flow of food, energy and the feeding relationship b/w them.

Food chain is just a sequence of organisms, in which each is food for the next.

Tertiary consumers → Decomposers
↑
Secondary consumers → Decomposers
↑
Primary consumers → Decomposers
↑
Producers → Decomposers

(food chain)

Grazing Food chain

→ grasslands & forest ecosystems follow GFC

Grass → Grasshopper → Frog → Snake → Hawk

green plants → Goat → Wolf → Lion

Detritus food chain

→ estuarine & mangrove leaf ecosystems follow DFC

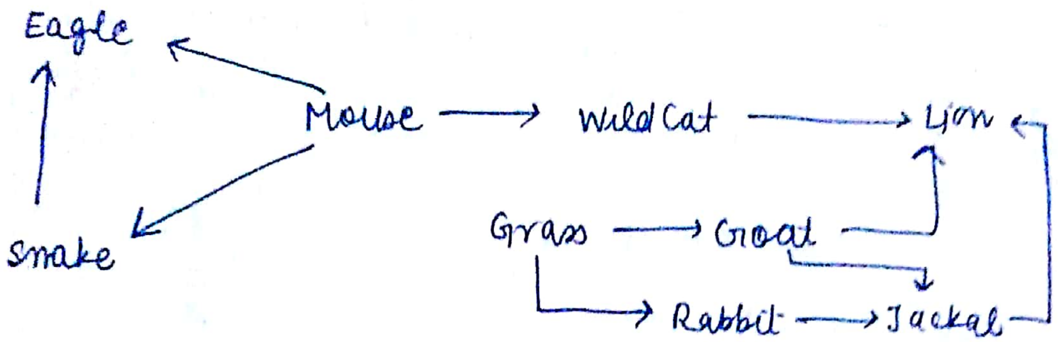
→ dead plants, dead animals & fallen leaves are consumed by detrivores & their predators

Dead plants → soil mites → Insects → Lizards

Dead organic matter → Bacteria → Protozoa → Rotifers

Food Web

The complex network of interconnected food chains is called a food web. (Food chains overlap because most consumers feed on multiple species & in turn, are fed upon by multiple other species)



(Food web)

Eg - a snake might feed on a mouse, a lizard or a frog. In turn, snake might be eaten by a bird or a badger

Q4 What do you understand by ecological succession?

Ecological succession is orderly changes in the composition or structure of an ecological community.

(i) Primary and secondary succession

a) When development of an area begins that has not been previously occupied by a community, the process is known as primary succession.
Eg - a lava flow, a newly formed lake

b) When community development is proceeding in an area from which a community was removed, it is called secondary succession.
Eg - it arises on cut-over forest, an abandoned crop

(ii) Seasonal & cyclic succession

These are periodic changes arising from fluctuating species interactions on recurring events.
eg. vegetation changes which are not dependent on disturbance

Causes of Plant succession

(i) changes in soil (pH, nutrients) by presence of microbes
(ii) changes in soil by external environmental influences (erosion, deposition of silt & clay)
(iii) changes caused by climate factors (temp, rainfall, global warming)

Q.5 Describe the structure of an ocean ecosystem stating its characteristic features.

Marine or ocean ecosystem

(A) Littoral zone (1): shoreline b/w land and open sea. Waves & tides have maximum effect in a littoral zone.

Region	Organism
(i) Rocky shore region	Starfish, barnacles, algae
(ii) Sandy shore region	Snails, clams
(iii) Bays	Algae

(B) Neritic zone (2): zone lies above continental shelf

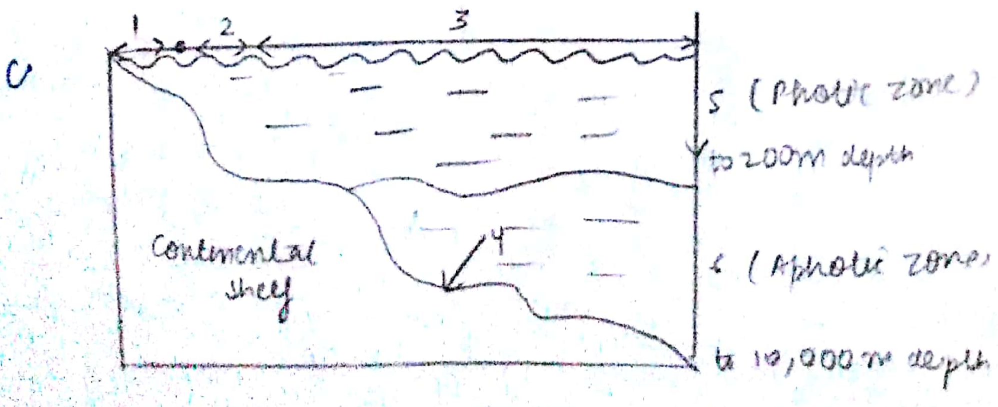
- This zone is rich in species since nutrients washed from land is found in this zone
- The productivity of this zone is high because sunlight can penetrate through this zone
- Pollution affects this zone. Zooplankton & phytoplankton are abundant here & support fishing grounds

(C) Pelagic zone (3)

- open sea constituting 90% of total ocean forms this zone.
- Phytoplanktons, zooplanktons, shrimps, jelly fish, fin, deepwater fishes & blue whales are found here
- organisms are below light penetration zone & totally depend on rain of detritus of upper regions for nutrition

(D) Benthic zone (4)

- floor of ocean constitutes this zone. It stretches from edge of continental shelf to deepest ocean trenches
- sponges, sea lilies, sea fans, snails, clams, starfish, sea cucumbers & sea urchins



Q6 Write a short note on Mangrove forests

Mangrove forests are mainly found along stretches of tropical ocean. They usually grow in places near quiet ocean waters. These forests oftens grow in shallow waters along bays, lagoons and river mouths. The thousands of stilt-like roots of a mangrove tree catch silt and sand, which piles up in the shore water. These roots slow down the current and help settle the silt. The mangroves, thus, help in building up dry land. out of total forest cover in india, less than one percent is mangrove forests

Q7 "The flow of energy is one way and continous in an ecosystem". Justify.

The flow of energy through the various components of the ecosystem is unidirectional and continuous. Unlike the nutrients, which move in a cyclic manner and are reused by the producers after flowing through the food chain, energy is not reused in the food chain.

All organisms require energy for growth, maintenance, reproduction, locomotion, etc. The flow of energy in an ecosystem follows the laws of thermodynamics.

(i) I Law : Energy can never be created or destroyed, but can be converted from one form to another.

(ii) II Law : Transformations of energy always result in some loss or dissipation of energy.